

CircuitikZ Figure Template

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About this template

This document renders an RC circuit (one resistor, one capacitor) using the `circuitikz` LaTeX package, inside a pandoc-managed LaTeX document. If `tlmgr install circuitikz` has not been run, this template will fail to compile — see the Notes section for the install hint.

Simple RC Low-Pass Filter

Figure 1 shows the canonical passive RC low-pass filter topology.

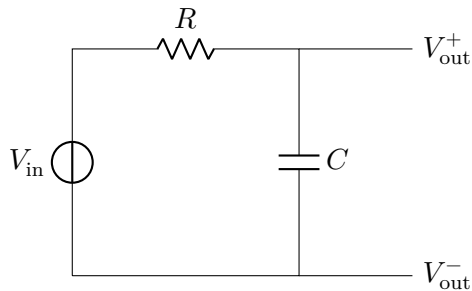


Figure 1: Passive RC low-pass filter with cutoff $f_c = 1/(2\pi RC)$.

Notes

- **Install requirement:** on a minimal TeX Live, run `tlmgr install circuitikz pgf` before first compile.
- `\ctikzset{bipoles/length=0.9cm}` in `header-includes` is a project-wide component-size preference; remove it to keep circuitikz defaults.
- `circuitikz` is TikZ-heavy; if compile time becomes an issue, try `--pdf-engine=lualatex` (CON-002 allows lualatex as a TikZ fallback).

Compile

```
pandoc circuitikz-figure.md \  
  --pdf-engine=xelatex \  
  -o circuitikz-figure.pdf
```